Quantifier proof worksheet

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Quantifier proof rules

$$\begin{array}{l|ll} i & \forall x S(x) & \\ & \vdots & \\ j & S(c) & \forall\,\text{Elim: } i \end{array}$$

where we obtain $S(c)$ from $S(x)$ by substituting all free occurrences of x with c

$$\begin{array}{l|ll} i & \boxed{c} & \\ & \vdots & \\ j & S(c) & \\ k & \forall x S(x) & \forall\,\text{Intro: } i\text{–}j \end{array}$$

where we obtain $S(c)$ from $S(x)$ by substituting all free occurrences of x in $S(x)$ with c

$$\begin{array}{l|ll} i & \exists x S(x) & \\ j & \boxed{c}\ S(c) & \\ & \vdots & \\ k & P & \\ l & P & \exists\,\text{Elim: } i,\ j\text{–}k \end{array}$$

where we obtain $S(c)$ from $S(x)$ by substituting all free occurrences of x in $S(x)$ with c

$$\begin{array}{l|ll} i & S(c) & \\ & \vdots & \\ j & \exists x S(x) & \exists\,\text{Intro: } i \end{array}$$

where we obtain $S(x)$ from $S(c)$ by replacing zero or more occurrences of c by x

Note: if you introduce a boxed constant $\boxed{c}$ in the premise of a subproof, you can only use c within that subproof.
Note: see the lecture slides for more details.

LPL 13.30

Question: Formally prove the following inference:

$$\begin{array}{l}\text{Likes}(\text{carl}, \text{max}) \\ \forall x[\exists y(\text{Likes}(y, x) \vee \text{Likes}(x, y)) \rightarrow \text{Likes}(x, x)] \\ \hline \exists x \text{Likes}(x, \text{carl})\end{array}$$

Thought process: first let's translate the argument into English, which hopefully gives us a better understanding.

Carl likes Max
Everybody who either is liked by someone or likes someone, likes themselves
Somebody likes Carl

Upon seeing this, it might start to make sense that this argument is true: after all, Carl likes Max (1st premise), so Carl likes someone, (so Carl either is liked by someone or likes someone), and thus, by the second premise, Carl likes himself. Thus, one can infer that somebody (namely, Carl) likes Carl, which is the conclusion.

Now, we only need to translate this reasoning to a formal proof ;)

So, following the informal reasoning above, let us try to construct the proof by first proving that Carl is liked by someone or likes someone ($\exists y(\text{Likes}(y, \text{carl}) \vee \text{Likes}(\text{carl}, y))$), then proving that Carl likes himself ($\text{Likes}(\text{carl}, \text{carl})$), finally proving that somebody likes Carl ($\exists x \text{Likes}(x, \text{carl})$).

Solution:

1	$\text{Likes}(\text{carl}, \text{max})$	
2	$\forall x[\exists y(\text{Likes}(y, x) \vee \text{Likes}(x, y)) \rightarrow \text{Likes}(x, x)]$	
3	$\text{Likes}(\text{max}, \text{carl}) \vee \text{Likes}(\text{carl}, \text{max})$	$\vee$ Intro: 1
4	$\exists y(\text{Likes}(y, \text{carl}) \vee \text{Likes}(\text{carl}, y))$	$\exists$ Intro: 3
5	$\exists y(\text{Likes}(y, \text{carl}) \vee \text{Likes}(\text{carl}, y)) \rightarrow \text{Likes}(\text{carl}, \text{carl})$	$\forall$ Elim: 2
6	$\text{Likes}(\text{carl}, \text{carl})$	$\rightarrow$ Elim: 5, 4
7	$\exists x \text{Likes}(x, \text{carl})$	$\exists$ Intro: 6

LPL 13.31

Question: Formally prove the following inference:

$$\begin{array}{l}\forall x \forall y[\text{Likes}(x, y) \rightarrow \text{Likes}(y, x)] \\ \exists x \forall y \text{Likes}(x, y) \\ \hline \forall x \exists y \text{Likes}(x, y)\end{array}$$

Thought process: Again, let us translate the argument into English:

Very informally: if you like someone, then they like you back
Somebody likes everybody
Everybody likes somebody

Let's try to first understand informally why this argument is true. One informal proof would be: *By the second premise, there is someone (let's call him Hector), who likes everybody. From the first premise, we can now infer that everybody likes Hector "back". Thus, everybody likes Hector, so everybody likes somebody (which is the conclusion).*

The difficulty with this proof is that you need two (nested) subproofs. One subproof is needed to introduce Hector as a boxed constant $\boxed{c}$ (and to eventually use $\exists$ Elim on), and the other subproof is needed to prove (with $\forall$ Intro) that everybody likes Hector.

First, as an intermediate step let's introduce Hector (or $\boxed{c}$) (proof on the left):

$\forall x \forall y[\text{Likes}(x,y) \to \text{Likes}(y,x)]$		$\forall x \forall y[\text{Likes}(x,y) \to \text{Likes}(y,x)]$	
$\exists x \forall y \text{Likes}(x,y)$		$\exists x \forall y \text{Likes}(x,y)$	
$\boxed{c}$ $\forall y \text{Likes}(c,y)$	$\rightsquigarrow$	$\boxed{c}$ $\forall y \text{Likes}(c,y)$	
...	$\rightsquigarrow$	$\boxed{d}$	
...	$\rightsquigarrow$	...	
$\forall x \exists y \text{Likes}(x,y)$		...	
$\forall x \exists y \text{Likes}(x,y)$		$\exists y \text{Likes}(d,y)$	
		$\forall x \exists y \text{Likes}(x,y)$	$\forall$ Intro: . . .
		$\forall x \exists y \text{Likes}(x,y)$	$\exists$ Elim: . . .

Reasoning backwards, it is reasonable to try to get the $\forall$ in the before-last line with $\forall$ Intro, for which another subproof is needed: see the proof on the right.

Now, all that is left to do, is to fill in the gap:

Full solution:

1	$\forall x \forall y[\text{Likes}(x,y) \to \text{Likes}(y,x)]$	
2	$\exists x \forall y \text{Likes}(x,y)$	
3	$\boxed{c}\ \forall y \text{Likes}(c,y)$	
4	$\boxed{d}$	
5	$\forall y[\text{Likes}(c,y) \to \text{Likes}(y,c)]$	$\forall$ Elim: 1
6	$\text{Likes}(c,d) \to \text{Likes}(d,c)$	$\forall$ Elim: 5
7	$\text{Likes}(c,d)$	$\forall$ Elim: 3
8	$\text{Likes}(d,c)$	$\to$ Elim: 6, 7
9	$\exists y \text{Likes}(d,y)$	$\exists$ Intro: 8
10	$\forall x \exists y \text{Likes}(x,y)$	$\forall$ Intro: 4–9
11	$\forall x \exists y \text{Likes}(x,y)$	$\exists$ Elim: 2, 3–10

LPL 13.52

Question: Formally prove the following inference:

$$\vdash \neg \exists x \forall y[E(x,y) \leftrightarrow \neg E(y,y)]$$

Thought process: instead of trying to translate this argument, here it might be easier to just start right away with the actual proof. We have to prove the negation of something; thus, let's aim for negation introduction and make a subproof that starts with $\exists x \forall y[E(x,y) \leftrightarrow \neg E(y,y)]$ and ends with $\bot$.

There is not much else we can do with the existential quantifier than to use $\exists$ Elim on it, meaning we should make another subproof, so that we get this:

$\exists x \forall y[E(x,y) \leftrightarrow \neg E(y,y)]$	
$\boxed{a}\ \forall y[E(a,y) \leftrightarrow \neg E(y,y)]$	
...	
...	
$\bot$	
$\bot$	$\exists$ Elim: . . .
$\neg \exists x \forall y[E(x,y) \leftrightarrow \neg E(y,y)]$	$\neg$ Intro: . . .

Observe that we can use $\forall$ Elim to get $E(a,a) \leftrightarrow \neg E(a,a)$, which is clearly a contradiction. The tricky part is to prove that it indeed leads to $\bot$, but it can be done the following way:

Solution:

1	$\exists x \forall y[E(x,y) \leftrightarrow \neg E(y,y)]$	
2	$\boxed{a}\ \forall y[E(a,y) \leftrightarrow \neg E(y,y)]$	
3	$E(a,a) \leftrightarrow \neg E(a,a)$	$\forall$ Elim: 2
4	$E(a,a)$	
5	$\neg E(a,a)$	$\leftrightarrow$ Elim: 3, 4
6	$\bot$	$\bot$ Intro: 4, 5
7	$\neg E(a,a)$	$\neg$ Intro: 4–6
8	$E(a,a)$	$\leftrightarrow$ Elim: 3, 7
9	$\bot$	$\bot$ Intro: 8, 7
10	$\bot$	$\exists$ Elim: 1, 2–9
11	$\neg\exists x \forall y[E(x,y) \leftrightarrow \neg E(y,y)]$	$\neg$ Intro: 1–10

LPL 13.39

Question: Formally prove the following inference:

$\forall x(\text{Cube}(x) \rightarrow \forall y(\text{Dodec}(y) \rightarrow \text{Larger}(x,y)))$
$\forall x(\text{Dodec}(x) \rightarrow \forall y(\text{Tet}(y) \rightarrow \text{Larger}(x,y)))$
$\exists x \text{Dodec}(x)$
$\forall x \forall y \forall z((\text{Larger}(x,y) \wedge \text{Larger}(y,z)) \rightarrow \text{Larger}(x,z))$
$\therefore \forall x(\text{Cube}(x) \rightarrow \forall y(\text{Tet}(y) \rightarrow \text{Larger}(x,y)))$

Thought process: again, let's give some meaning to this argument:

All cubes are larger than all dodecahedra
All dodecahedra are larger than all tetrahedra
There exists a dodecahedron
If x is larger than y and y larger than z, then x must be larger than z (for all x, y, z)
$\therefore$ All cubes are larger than all tetrahedra

This argument may seem hard to prove, but luckily, it is the perfect example of a "working backwards" proof.

We want to prove $\forall x(\text{Cube}(x) \rightarrow \forall y(\text{Tet}(y) \rightarrow \text{Larger}(x,y)))$. It would be reasonable to guess that this universal quantifier in the front has been introduced using $\forall$ Intro, so we should make a subproof that starts with $\boxed{a}$ and ends with $\text{Cube}(a) \rightarrow \forall y(\text{Tet}(y) \rightarrow \text{Larger}(a,y))$. Here, the main connective is an implication, suggesting that we could make a subproof that starts with $\text{Cube}(a)$

and ends with $\forall y(\text{Tet}(y) \to \text{Larger}(a, y))$. In this last sentence, the main connective is $\forall$ again, so we can continue working backwards, to get this:

$\forall x(\text{Cube}(x) \to \forall y(\text{Dodec}(y) \to \text{Larger}(x, y)))$
$\forall x(\text{Dodec}(x) \to \forall y(\text{Tet}(y) \to \text{Larger}(x, y)))$
$\exists x \text{Dodec}(x)$
$\forall x \forall y \forall z((\text{Larger}(x, y) \wedge \text{Larger}(y, z)) \to \text{Larger}(x, z))$

$\boxed{a}$
$\text{Cube}(a)$

$\boxed{b}$

$\text{Tet}(b)$

$\ldots$
$\ldots$
$\text{Larger}(a, b)$
$\text{Tet}(b) \to \text{Larger}(a, b)$ $\to$ Intro: . . .
$\forall y(\text{Tet}(y) \to \text{Larger}(a, y))$ $\forall$ Intro: . . .
$\text{Cube}(a) \to \forall y(\text{Tet}(y) \to \text{Larger}(a, y))$ $\to$ Intro: . . .
$\forall x(\text{Cube}(x) \to \forall y(\text{Tet}(y) \to \text{Larger}(x, y)))$ $\forall$ Intro: . . .

Now, we have to fill in the part in the middle. The informal reasoning would be:

a is a cube and b is a tetrahedron. By the third premise, there exists a dodecahedron (let's call it c). Using the first premise, we can infer that the cube a is larger than the dodecahedron c, and using the second premise we can infer that the dodecahedron c is larger than the tetrahedron b. Using these two facts and the fourth premise, it becomes clear that a is larger than b, which is what we need to reach.

This idea can be made formal by making another subproof in which we introduce $\boxed{c}$, followed by $\exists$ Elim. Inside that subproof, it turns out we just need many $\forall$ Elim and $\to$ Elim in order to prove what is needed:

Solution:

1	$\forall x(\text{Cube}(x) \to \forall y(\text{Dodec}(y) \to \text{Larger}(x, y)))$	
2	$\forall x(\text{Dodec}(x) \to \forall y(\text{Tet}(y) \to \text{Larger}(x, y)))$	
3	$\exists x\text{Dodec}(x)$	
4	$\forall x\forall y\forall z((\text{Larger}(x, y) \land \text{Larger}(y, z)) \to \text{Larger}(x, z))$	
5	$\boxed{a}$	
6	$\text{Cube}(a)$	
7	$\boxed{b}$	
8	$\text{Tet}(b)$	
9	$\boxed{c}\ \text{Dodec}(c)$	
10	$\text{Cube}(a) \to \forall y(\text{Dodec}(y) \to \text{Larger}(a, y))$	$\forall$ Elim: 1
11	$\forall y(\text{Dodec}(y) \to \text{Larger}(a, y))$	$\to$ Elim: 9, 6
12	$\text{Dodec}(c) \to \text{Larger}(a, c)$	$\forall$ Elim: 11
13	$\text{Larger}(a, c)$	$\to$ Elim: 12, 9
14	$\text{Dodec}(c) \to \forall y(\text{Tet}(y) \to \text{Larger}(c, y))$	$\forall$ Elim: 2
15	$\forall y(\text{Tet}(y) \to \text{Larger}(c, y))$	$\to$ Elim: 14, 9
16	$\text{Tet}(b) \to \text{Larger}(c, b)$	$\forall$ Elim: 15
17	$\text{Larger}(c, b)$	$\to$ Elim: 16, 8
18	$\forall y\forall z((\text{Larger}(a, y) \land \text{Larger}(y, z)) \to \text{Larger}(a, z))$	$\forall$ Elim: 4
19	$\forall z((\text{Larger}(a, c) \land \text{Larger}(c, z)) \to \text{Larger}(a, z))$	$\forall$ Elim: 18
20	$(\text{Larger}(a, c) \land \text{Larger}(c, b)) \to \text{Larger}(a, b)$	$\forall$ Elim: 19
21	$\text{Larger}(a, c) \land \text{Larger}(c, b)$	$\land$ Intro: 13, 17
22	$\text{Larger}(a, b)$	$\to$ Elim: 20, 21
23	$\text{Larger}(a, b)$	$\exists$ Elim: 3, 9–22
24	$\text{Tet}(b) \to \text{Larger}(a, b)$	$\to$ Intro: 8–23
25	$\forall y(\text{Tet}(y) \to \text{Larger}(a, y))$	$\forall$ Intro: 7–24
26	$\text{Cube}(a) \to \forall y(\text{Tet}(y) \to \text{Larger}(a, y))$	$\to$ Intro: 6–25
27	$\forall x(\text{Cube}(x) \to \forall y(\text{Tet}(y) \to \text{Larger}(x, y)))$	$\forall$ Intro: 5–26